

King Jin

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EDUCATION

The University of Edinburgh

Bachelor of Science in Artificial Intelligence

Edinburgh, UK

Sep 2024 – May 2028

- First Year average: 90/100
- Relevant Coursework: Linear Algebra, Discrete Maths and Probability, Several Variable Calculus and DEs, Object Oriented Programming, Data Science, Algorithms and Data Structures, Reasoning and Agents

EXPERIENCE

Software Engineer – Perception

Edinburgh University Formula Student (EUFS)

Edinburgh, UK

Sep 2025 – Present

- Developed autonomous racing vision pipelines for cone detection, combining LiDAR and camera data through sensor fusion for FSUK competitions.
- Building a camera inference pipeline for the ZED stereo camera, processing frames through detection models with **ONNX Runtime** and **TensorRT** for optimized real-time inference.
- Created and labeled a comprehensive custom cone dataset using **CVAT**, establishing a reusable annotation and data pipeline for the perception team.
- Fine-tuned **YOLOv11** and **RF-DETR** on the custom dataset, benchmarking model performance across detection and segmentation metrics.

SparrowNet: An Ensemble Approach to Hieroglyph Character Recognition

Researcher, Southeast University Summer AI Program

Nanjing, China

Jul 2023 – Sep 2023

- Proposed **SparrowNet**, a novel ensemble neural network combining lightweight ResNet variants with 1x1 convolutions.
- Benchmarked SparrowNet against ResNet18 and GoogLeNet V2 on an open-source Egyptian hieroglyph dataset, analyzing efficiency and accuracy across multiple metrics.
- Produced a formal research paper outlining methodology and results.

PROJECTS

CamerAI | NCNN, Android, OpenCV, Python

Jan 2026 – Present

- Building a real-time, on-device AI photography assistant for Android (Snapdragon 8 Gen 3) using NCNN for efficient neural network inference.
- Fused two research architectures: **GAIC** (CVPR 2019) for efficient composition proposal generation and **SAMP-Net** (BMVC 2021) for aesthetic quality evaluation.
- Implemented real-time on-screen arrow guidance to direct users toward optimal framing and composition.

Lets-break-the-information-gap | FastAPI, PostgreSQL, React, Gemini API, Docker

Jun 2025 – Present

- Developed a web app that converts RSS feeds into personalized, multi-speaker podcasts.
- Integrated Gemini API to generate scripts and automate podcast creation.
- Implemented full-stack CI/CD pipeline with containerized deployment.

Home Server Setup | Docker, Linux

Dec 2024 – Present

- Repurposed an HP-Elitedesk-800-G4-SFF to host Ubuntu Server running Docker containers (Nextcloud, Jellyfin).
- Deployed **OpenClaw** as a self-hosted personal AI assistant.
- Used for deploying and testing personal frontend/backend projects.

SKILLS

Languages: Chinese (Native), English (Fluent)

Programming: Python, Java, Haskell, SQL, Markdown, LaTeX, HTML/CSS, JavaScript, C++

Frameworks: React, FastAPI, PyTorch, Electron, ONNX, NCNN, TensorRT, OpenCV, Hugging Face

Developer Tools: Git, Docker, Unix/Linux, ROS2, CVAT, Claude Code

Other Skills: Containerization, Deep Learning, NumPy, Pandas, Matplotlib, vLLM, [Learn more..](#)